

However, all this research doesn't seem to have made much of an impact in preventing the spread of COVID-19. One might be inclined to question the efficacy of AI itself at this point. However, being an AI researcher who has closely followed the developments in the AI community during this crisis, I have reasons to disagree with that notion.

On January 6, 2020, the Centre for Disease Control (CDC) of the USA issued a travel advisory for people travelling to China regarding the COVID-19. This was followed by the WHO's advisory on January 9. However, an AI powered health monitoring system based in Canada named BlueDot had issued an advisory regarding this outbreak on December 31, 2019, a full week before any government body or health organisation was able to predict the outbreak. What makes things more interesting is that the predictions that BlueDot made regarding the spread of the virus turned out to be very accurate. BlueDot uses natural language processing and machine learning to scour thousands of news reports across the globe, as well as airline ticketing information, to correlate and predict the spread of an epidemic/pandemic. This is a much superior approach as compared to trusting the official information shared by a country's government, which may not always be accurate or provide a complete picture on the ground.

The example of BlueDot proves that it is not the efficacy of AI that is to be blamed but the delay in our response to the pandemic. Simply put, we were slow on pulling the trigger on this one. Nobody expected COVID-19 to spread as fast as it did and have such a major impact on the entire world. So, we were slow to act. Not enough of an effort was made during the months of

January and February. By the time people started to realise the potential of the threat that COVID-19 posed, March had arrived and the disease was in full swing, causing countrywide shutdowns and chaos.

How is AI currently helping with the COVID-19 pandemic?

While AI may not have been very useful in preventing the pandemic from occurring, it is certainly playing a role in containing the pandemic and helping medical teams in treating the infected patients.

- Facebook is currently working with researchers at Harvard University's School of Public Health and the National Tsing Hua University, in Taiwan, sharing anonymised data about people's movements and high-resolution population density maps, which help them forecast the spread of the virus.
- Google is using machine learning and AI to fish out the misinformation regarding COVID-19 on its platforms.
- Exscientia, a UK based pharmaceutical company, is using AI to develop a drug to fight against the spread of COVID-19.
- Deep Mind, a Google owned AI based company, is using its AlphaFold system to release structure predictions of several proteins associated with the virus.
- Companies such as BlueDot and Metabiota have been key players in predicting the growth of the pandemic, which is vital information for both the governments and the medical staff.
- Alibaba Cloud, DAMO Academy, and DingTalk have joined forces to launch a series of AI technologies and cloud based solutions to support companies and organisations worldwide in the fight against COVID-19.
- Researchers in the field of AI from the Renmin Hospital of the Wuhan

University have developed an AI solution that detects COVID-19 in patients with 95 per cent accuracy, by analysing the CT scan of a patient's chest.

- The Israel based medtech company, Nanox, has developed a mobile digital X-ray system that uses AI cloud based software to diagnose infections and help prevent epidemic outbreaks.

Open source projects in the fight against COVID-19

While the big tech companies and research institutes have been doing their part, many open source projects have also been started, using the power of AI in the fight against the COVID-19 pandemic.

CHIME, a COVID-19 hospital impact model, has been made open source by the Predictive Health Care team at Penn Medicine. CHIME uses SIR modelling to predict the number of people that can be infected in a closed population. This tool is proving to be very useful for hospitals to predict and plan the number of beds required to treat the patients.

NEXTSTRAIN is a real-time tracking program for pathogen evolution which is currently being used to monitor COVID-19. This technology, built by the Global Initiative on Sharing All Influenza Data (GISAID), had previously proven to be very useful in the fight against Ebola.

In India, CoronaSafe is an open source initiative launched with the support of the Kerala government to combat COVID-19 on multiple fronts. The Corona Literacy Mission, Corona Care Centres and real-time updates of COVID-19 cases in India have been made available on this platform. A wide array of scientists and engineers from multiple disciplines, including AI, are participating in this project.

The OpenCovid-19 initiative launched by Just One Giant Lab (JOGL) is working on building